

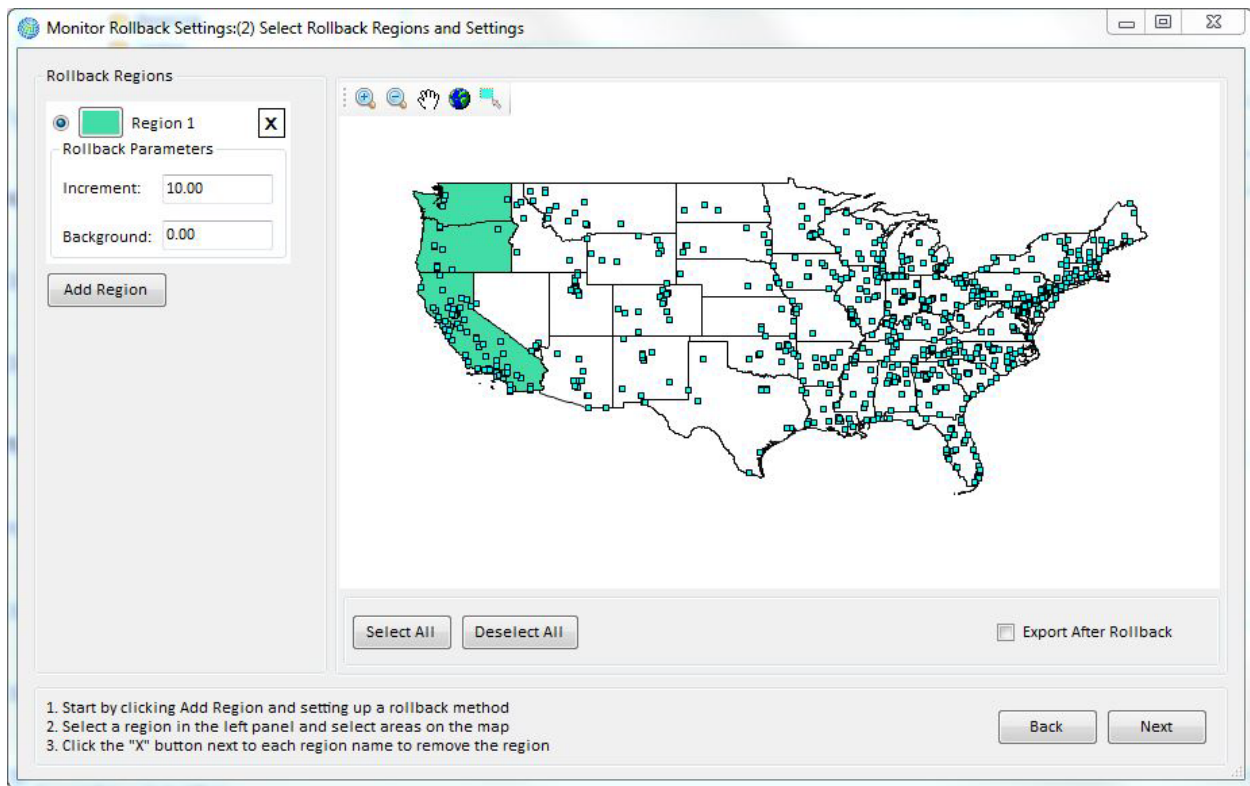
Note that there is an **Advanced** button that lets you select the **Maximum Neighbor Distance (in km)**, **Maximum Relative Neighbor Distance**, and **Weighting Approach**. The specific availability of advanced features depends on the interpolation method that you choose. The **Advanced Options** window is described in more detail in Section 5.2.4. You can view a map of the inputs to the rollback grids that you are creating, as well as the grids themselves by accessing the **Monitor Map**. To do this, click on the **Advanced** button, then the **Custom Monitor Filtering** button, and then the **Map** button (on the **Filter Monitors** window).

5.3.2 Example: Combining Three Rollback Approaches in Different Regions

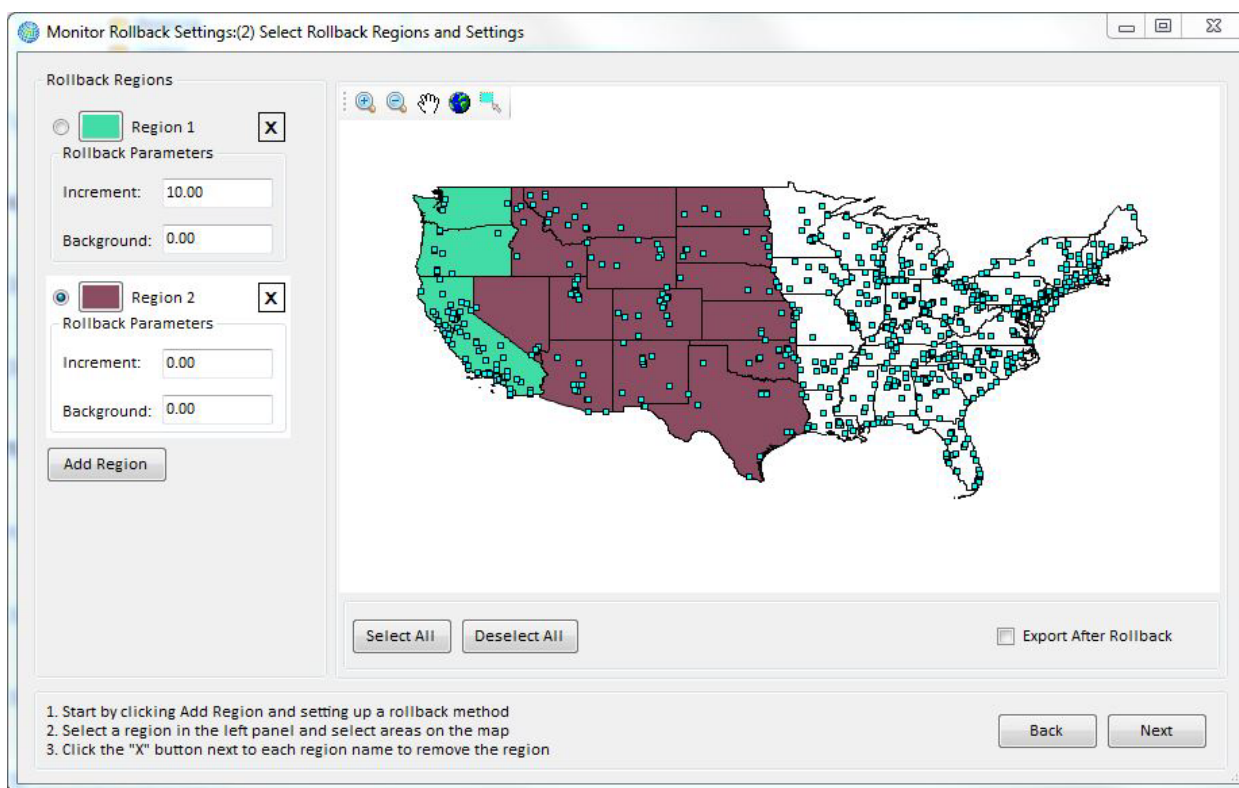
BenMAP-CE allows you to use different approaches to rolling back air quality data among regions. In this example, we'll use the *United States* setup to combine the three rollback types: *Percentage Rollback*, *Incremental Rollback*, and *Rollback to a Standard*.

Start by clicking on the **Pollutant** button in the left-hand pane of the main screen, and choosing *Ozone*. Next, click **Baseline** under **Source of Air Quality Data**. Select *Monitor Rollback* as the **Grid Creation Method**. On the **Monitor Rollback: (1) Select Monitors** window, select the **Rollback Grid Type** (*State*), **Monitor Dataset** (*EPA Standard Monitors O3*) and **Monitor Library Year** (*2000*), and click **Next**.

On the **Monitor Rollback Settings: (2) Select Rollback Regions and Settings** window, click the **Add Region** button and select the *Percentage Rollback* method. Click **OK**. Enter *10* for the **Percent**. In the previous example, we included only one state in the rollback region. In this example we want to create three regions. Click on the three western-most states to add them to the first region. The states you have added to the region will fill in, as in the picture below.



To add states with a second type of rollback, click on the **Add Region** button, choose the rollback type, and then click on the states to include in this second region, which BenMAP-CE denotes as Region 2. In this example, we have chosen an *Incremental Rollback* with an **Increment** of 5 and a **Background** of 0, and applied it to the 14 next western-most states.



The map now depicts two rollback regions. We can toggle back and forth between each region by clicking on the button on the legend on the left side of the map. Any states that have not yet been included in a region may be added to an existing region, or we may create one or more regions for these remaining states. Note that once states have been included in a rollback region, they cannot be included in a different rollback region. In our example, the three western-most states are highlighted in gray.

If you want to add or remove states from a defined region, make sure you select the appropriate region by clicking on the button to the left of that region before clicking on the map to select or deselect the state(s).

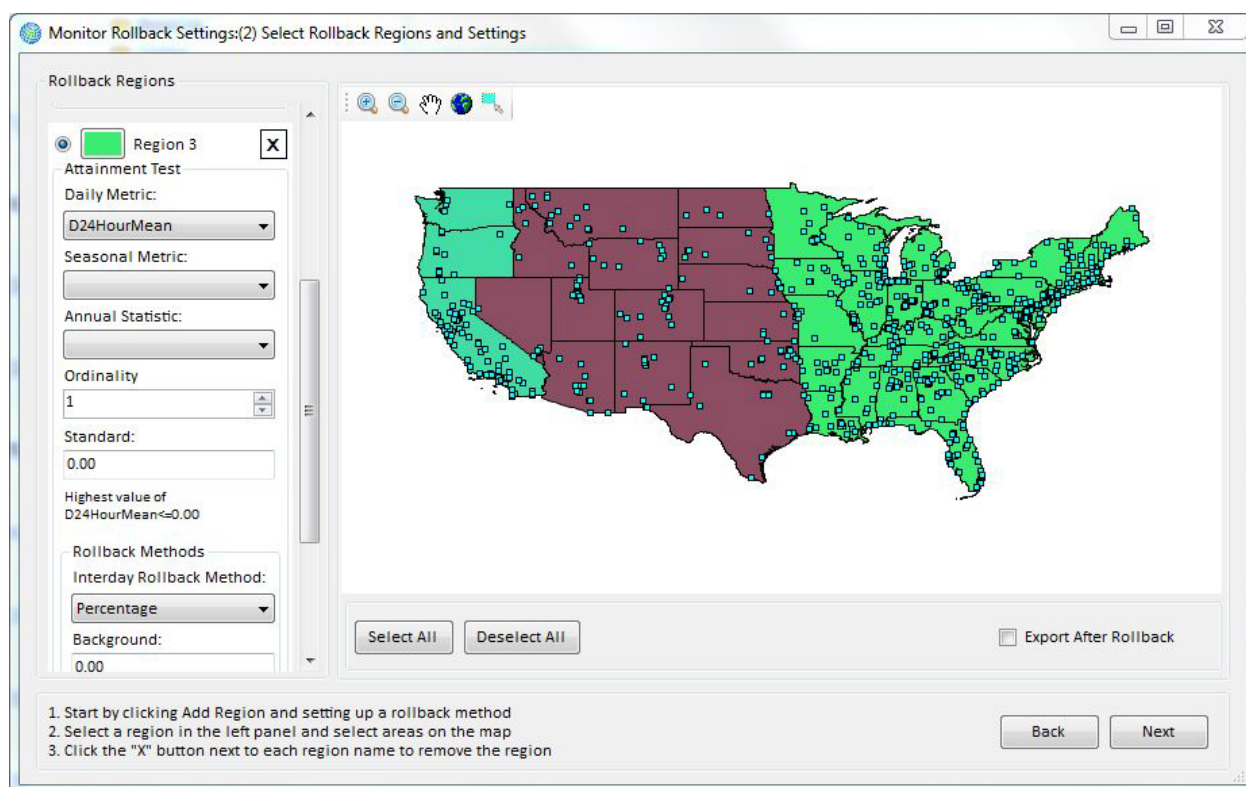
To add a third rollback type covering the rest of the states, click again on the **Add Region** button, and then choose the rollback type. However, instead of individually choosing the states, simply click the **Select All** button. This will select all of the states that are not yet included in a region, and these remaining states will now become Region 3.

In this third region, we have chosen a *Rollback to a Standard*, which involves two groups of parameters - those associated with the **Attainment Test**, which determines whether a monitor is in attainment (meets the standard), and those associated with the **Rollback Methods**, which are used to bring out-of-attainment monitors into attainment.

The **Attainment Test** parameters are **Daily Metric**, **Seasonal Metric**, **Annual Statistic**, **Ordinality**, and **Standard**. (Note: You will need to use the scroll bar to view more detail on the left side of the screen.)

In this step BenMAP-CE calculates the metric to be used to determine whether a monitor's values must be rolled back and, if so, how much (e.g., if **Metric** is *D8HourMax*, BenMAP-CE calculates the 8-hour daily maximum for each day at each monitor).

A monitor is considered in attainment if the n^{th} highest value of a daily metric specified by **Metric** is at or below the value specified by **Standard**, where n is the value specified by **Ordinality**. For example, if **Metric** is *D8HourMax*, **Ordinality** is 4, and **Standard** is 85, a monitor will be considered in attainment if the fourth highest value of the eight-hour daily maximum is at or below 85 ppb.



The **Attainment Test** can also be used for seasonal metrics (by choosing previously defined seasonal metrics from the drop-down list below **Seasonal Metric**), as well as for annual metrics (by using the drop-down list below **Annual Statistic**). For example, if you want the annual mean ozone level to stay below 60 ppb, then you would choose the daily 24 hour mean (*D24HourMean*) from the drop-down list below **Daily Metric**, choose *Mean* from the drop-down list below **Annual Statistic**, and set the **Standard** to 60. (Note that in this case **Ordinality** cannot be chosen because there is only a single annual value.)